

SCREENING-GRADE WATERMARK. Not an engineering design, not a field-verified parcel survey, not an observed intervention trial.

Purpose: screening-grade, funder-oriented preventive action plan (auto-filled from artifacts)

Portfolio status: counterfactual simulation - deterministic candidate suitability plus literature-based intervention effects; field verification required

Evidence labels: observed data | model output | literature assumption | counterfactual simulation

#1. Decision context

The early-period 1-in-100-year daily rainfall depth now has a fitted recurrence of 14.28 years in the late-period sample. [model output from observed NOAA ISD]

Annual-max trend: 5.201 mm/decade. [model output]

Lake-area series summary: not available. [observed/derived inventory data]

Independent check (ERA5-Land at the same station coordinates): replicates (13.42-yr vs ISD 7.75-yr; annmax Pearson $r=0.432$, mean bias 28.09 mm). quasi-independent: reanalysis assimilates some gauges, but it is a different measurement and modeling system from raw ISD station records [model output on reanalysis]

#2. Evidence register

- Observed data: 1 NOAA stations / 45 station-years; UNOSAT Sentinel-1 flood extent for None; observed flood area not available km².
- Model output: GEV return levels {'2': 36.407, '5': 53.87, '10': 71.947, '25': 106.451, '50': 144.409, '100': 197.22} with bootstrap intervals {'2': [33.247, 40.912], '5': [44.305, 64.93], '10': [54.709, 96.766], '25': [69.342, 169.086], '50': [83.578, 287.942], '100': [96.386, 463.762]}; landslide rainfall classifier, AUC out-of-sample - not a Caine-style I-D threshold with held-out AUC None on 0 events; modeled flood area 493.46 km².
- Hazard method: HAND valley mask only. This is a local-min HAND proxy calibrated on this event, not Whitebox HAND or an independently validated hydrodynamic model.
- Calibration-event metrics: Calibration-event metrics are unavailable; this plan does not invent a score..
- Validation table:

Event	Role	CSI	POD	FAR
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calibration	in-sample calibration	not available	not available	not available

Baselines (same 2024 domain): not computed. Permanent-water rule: permanent water excluded where available.

- Counterfactual simulation: Counterfactual exposure 0.0 -> 0.0 (0.0% reduction; simulation using observed-flood fraction x population, not observed outcomes or unique lives).
- Tail cross-check: POT/GPD return levels {'2': 42.192, '5': 53.748, '10': 63.56, '25': 78.127, '50': 90.495, '100': 104.161} [model output; different sample from the GEV headline].
- Data gap: GPM IMERG V07 is on NASA Earthdata (login). Not fused; headline is ISD GEV only.

#3. Prioritized preventive interventions

Priority follows the optimizer's selected order. Coordinates are candidate centroids in EPSG:4326 and require site verification before procurement.

Priority	Measure ID	Measure	Coordinates (lat, lon)	Budget USD	Annual expected people-risk avoided	Risk driver	Suitability evidence	Evidence basis
---: ---: ---: ---: ---: ---: ---: ---: ---:	1	blr_p_0275	riverbank_bio	12.97000, 77.56000	6,000	714.9	flood	candidate inputs: slope 1.4 degrees, land cover urban Bioengineered riverbank protection (live crib walls, brush layering) per ICIMOD/DSCWM manuals.
2	blr_p_0274	riverbank_bio	12.97000, 77.54000	6,000	683.9	flood	candidate inputs: slope 1.4 degrees, land cover urban Bioengineered riverbank protection (live crib walls, brush layering) per ICIMOD/DSCWM manuals.	
3	blr_p_0253	riverbank_bio	12.99000, 77.60000	6,000	664.3	flood	candidate inputs: slope 1.4 degrees, land cover urban Bioengineered riverbank protection (live crib walls, brush layering) per ICIMOD/DSCWM manuals.	
4	blr_p_0205	riverbank_bio	13.03000, 77.60000	6,000	636.8	flood	candidate inputs: slope 1.3 degrees, land cover urban Bioengineered riverbank protection (live crib walls, brush layering) per ICIMOD/DSCWM manuals.	
5	blr_p_0226	riverbank_bio	13.01000, 77.54000	6,000	591.3	flood	candidate inputs: slope 2.0 degrees, land cover urban Bioengineered riverbank protection (live crib walls, brush layering) per ICIMOD/DSCWM manuals.	
6	blr_p_0251	riverbank_bio	12.99000, 77.56000	6,000	580.4	flood	candidate inputs: slope 1.5 degrees, land cover urban Bioengineered riverbank protection (live crib walls, brush layering) per ICIMOD/DSCWM manuals.	
7	blr_p_0203	riverbank_bio	13.03000, 77.56000	6,000	547.2	flood	candidate inputs: slope 1.1 degrees, land cover urban Bioengineered riverbank protection (live crib walls, brush layering) per ICIMOD/DSCWM manuals.	
8	blr_p_0252	riverbank_bio	12.99000, 77.58000	6,000	544.8	flood	candidate inputs: slope 1.4 degrees, land cover urban Bioengineered riverbank protection (live crib walls, brush layering) per ICIMOD/DSCWM manuals.	
9	blr_p_0224	riverbank_bio	13.01000, 77.50000	6,000	519.9	flood	candidate inputs: slope 1.6 degrees, land cover urban Bioengineered riverbank protection (live crib walls, brush layering) per ICIMOD/DSCWM manuals.	
10	blr_p_0204	riverbank_bio	13.03000, 77.58000	6,000	497.7	flood	candidate inputs: slope 1.3 degrees, land cover urban Bioengineered riverbank protection (live crib walls, brush layering) per ICIMOD/DSCWM manuals.	
11	blr_p_0206	riverbank_bio	13.03000, 77.62000	6,000	476.5	flood	candidate inputs: slope 1.1 degrees, land cover urban Bioengineered riverbank protection (live crib walls, brush layering) per ICIMOD/DSCWM manuals.	
12	blr_p_0273	riverbank_bio	12.97000, 77.52000	6,000	444.5	flood	candidate inputs: slope 2.4 degrees, land cover urban Bioengineered riverbank protection (live crib walls, brush layering) per ICIMOD/DSCWM manuals.	

#4. Budget and modeled outputs

Measure	Sites	Modeled spend USD	Annual expected people-risk avoided	
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riverbank_bio	21	126,000	9046.4	

- Budget ceiling: \$2,000,000; modeled spend: \$1,998,000; mode: expected.
- Annual expected people-risk avoided: 54253.6 [model output; not unique people or observed lives saved].
- Ten-year carbon: 9,780 tCO₂ [model output from literature factors].
- Annual livelihood-income potential: \$72,900 [model output from per-hectare literature assumptions; not measured income, jobs, wages, or households reached].
- Optimality: Greedy gap vs relaxed knapsack upper bound: 0.00% (bound \$272,485,877). Upper bound ignores per-cell overlap capping, so the true gap is smaller.

#5. Assumptions and evidence basis

- vetiver_slope - cost 5,000 USD/ha; effect fraction 0.35; carbon 12.0 tCO₂/ha/10yr; income 600 USD/ha/yr. Source: Vetiver hedgerows cut slope erosion >90% (Truong et al.); cost/income from ICIMOD bioengineering field costings. [literature assumption]
- bamboo_slope - cost 4,000 USD/ha; effect fraction 0.30; carbon 45.0 tCO₂/ha/10yr; income 850 USD/ha/yr. Source: Bamboo root reinforcement for shallow landslides; bamboo is an established Nepali rural cash crop (INBAR). [literature assumption]
- floodplain_restore - cost 8,000 USD/ha; effect fraction 0.30; carbon 35.0 tCO₂/ha/10yr; income 300 USD/ha/yr. Source: NbS flood-depth reduction ~15-40% (UNEP); floodplain reconnection costs from World Bank NbS catalogue. [literature assumption]
- wetland_restore - cost 12,000 USD/ha; effect fraction 0.35; carbon 60.0 tCO₂/ha/10yr; income 400 USD/ha/yr. Source: Wetland attenuation of peak flows; high soil-carbon uptake (Ramsar / IPCC wetlands supplement). [literature assumption]
- afforestation - cost 3,000 USD/ha; effect fraction 0.20; carbon 80.0 tCO₂/ha/10yr; income 250 USD/ha/yr. Source: Catchment afforestation for slope stability and infiltration; sequestration from IPCC AFOLU defaults. [literature assumption]
- riverbank_bio - cost 6,000 USD/ha; effect fraction 0.25; carbon 20.0 tCO₂/ha/10yr; income 500 USD/ha/yr. Source: Bioengineered riverbank protection (live crib walls, brush layering) per ICIMOD/DSCWM manuals. [literature assumption]

The optimizer applies marginal greedy triple-return per dollar, per-cell EAL capping, and Monte-Carlo climate multipliers derived from the GEV 100-year confidence interval. [model design assumption]

Equity weighting (verbatim from hazard/proof.py _grid_features): rural = landcover in cropland/grass/shrub AND no critical_assets; dense = population at or above the 70th percentile of cell pops; low_income_score = 1.0 if (rural and dense) else (0.55 if rural else 0.15); equity_weight = 1.0 + 0.5 * low_income_score (range 1.075-1.5x). eal_people is multiplied by equity_weight, so low-income rural exposure ranks higher in expected-loss. [model design assumption]

No causal percentage is assigned to government, communities, or households. Implementation roles require local governance and community confirmation.

#6. Monitoring and verification

- Rainfall signal: append observed NOAA ISD records annually; re-fit GEV and publish sample size, confidence interval, and any changed recurrence estimate.
- Flood extent: after qualifying events, derive observed Sentinel-1 extent and report POD, FAR, and CSI separately from the calibration event.
- Intervention delivery: verify each parcel centroid, boundary, measure type, area, and installation date through field sign-off plus dated optical imagery.

- Carbon: report planted/restored area and survival first; convert to tCO2 only with the stated factor and uncertainty. Ground-sample biomass before claiming credits.
- Risk performance: compare post-event observed exposure with a pre-registered baseline; do not present the counterfactual simulation as an observed outcome.
- Livelihoods: measure participant payments or income directly before making beneficiary, household, job, or wage claims.

#7. Limitations

- The flood surface is a screening-grade local-min HAND proxy calibrated on the 27 Sep 2024 event. Its event-fit CSI is low and is not independent validation.
- Hazard depths, EAL, avoided people-risk, carbon, income, and the optimized ranking are model outputs; field feasibility, tenure, ecological safeguards, and engineering design remain unverified.
- Candidate centroids are screening locations, not surveyed parcel boundaries.
- Literature effect sizes and unit costs may not transfer to local implementation conditions.
- The counterfactual is a simulation, not a prediction or measured causal effect.
- IMERG was not fused, and the HMA-pooled late GEV did not identify the Nepal-adjacent headline shift.